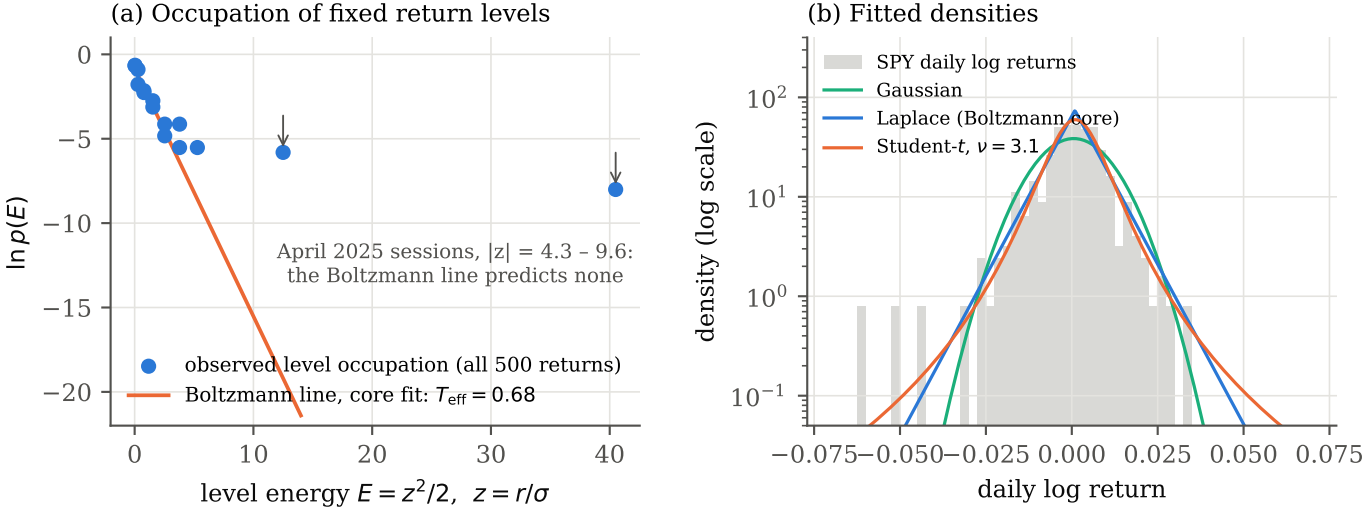
(a) Occupation of fixed return levels
0
−5
−10
−15
−20
ln p(E)
April 2025 sessions, |z| = 4.3 – 9.6:
the Boltzmann line predicts none
observed level occupation (all 500 returns)
Boltzmann line, core fit: T_eff = 0.68
0
10
20
30
40
level energy E = z²/2, z = r/σ
(b) Fitted densities
SPY daily log returns
Gaussian
Laplace (Boltzmann core)
Student-t, ν = 3.1
density (log scale)
10²
10¹
10⁰
10⁻¹
−0.075
−0.050
−0.025
0.000
0.025
0.050
0.075
daily log return